

① Bonding & Structure

Ionic Bond (e^- transferred) $\sim \frac{e^2}{4\pi\epsilon_0 R}$

$\Delta E = \text{Electron Affinity} + \text{Cohesive Energy}$
(required) $- \text{Ionisation energy} > 0$

Covalent Bond (e^- shared,
low E from delocalization)

Molecules (Van der Waals, alignment
of e^- dipoles)

$$\underline{E} = \frac{P_1}{4\pi\epsilon_0 r^3} \hat{r}$$

$$U = - \underbrace{P_2}_{\uparrow \chi_e E} \cdot \underline{E} = - \frac{P_1^2 \chi_e}{(4\pi\epsilon_0)^2} \frac{1}{r^6}$$

$$F = -\nabla U \propto r^{-7}$$

Heat Capacity

$$C_{V,P} = \left(\frac{\partial U}{\partial T} \right)_{V,P}$$

Compressibility

$$\beta_s = -\frac{1}{V} \frac{\partial V}{\partial P} = -\frac{1}{L} \frac{\partial L}{\partial F}$$

Classical: DuLong-Petit $U = 3Nk_B T$
 $C = 3Nk_B$

Einstein Model

QHO of
same freq ω .

$$E_n = (n + \frac{1}{2}) \hbar \omega$$

$$Z_1 = \sum_{n=0}^{\infty} e^{-\beta \hbar \omega (n + \frac{1}{2})} = \frac{e^{-\frac{1}{2} \beta \hbar \omega}}{1 - e^{-\beta \hbar \omega}}$$

$$U = - \frac{\partial}{\partial \beta} \ln Z \quad (Z = Z_1^{3N})$$

$$C_V = \frac{\partial U}{\partial T} = 3Nk_B (\beta \hbar \omega)^2 \frac{e^{\beta \hbar \omega}}{(e^{\beta \hbar \omega} - 1)^2}$$

low T: $\beta \hbar \omega \gg 1$, $\rightarrow C_V \approx 3Nk_B (\beta \hbar \omega)^2 e^{-\beta \hbar \omega}$

high T: $\beta \hbar \omega \ll 1$, $\rightarrow C_V \approx 3Nk_B$

Debye Model

phonons: discrete quanta of vibration.

① Dispersion by $\omega = c_s k$

② Bose-Einstein statistics $\bar{n}_i = \frac{1}{e^{\beta \hbar \omega_i} - 1}$

Density of states: 1 longitudinal, 2 transverse

$$g(k) dk = 3 \frac{V}{(2\pi)^3} 4\pi k^2 dk = \frac{3V k^2}{2\pi^2} dk$$

$$\int_0^{\omega_D} d\omega g(\omega) = 3N$$

Cutoff freq.

$$\omega_D = C_s (6n\pi^2)^{1/3}$$

$$U = \int_0^{\omega_D} d\omega \bar{n}_i g(\omega) \hbar \omega$$

[1] can set $C_v \propto T^3$ for low T .
($T \ll T_D$)

• Debye freq & temp: $\hbar \omega_D = k_B T_D$

[2] here $U = 3N k_B T$ for high T
($e^x \approx 1+x$ approx.) : ($T \gg T_D$)

Metals

Drude transport

$$\frac{d\mathbf{p}}{dt} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) - \frac{\mathbf{p}}{\tau}$$

using $\mathbf{p} = m\mathbf{v}$ & $\mathbf{j} = -ne\mathbf{v}$

steady state: $0 = -e\mathbf{E} + \frac{\mathbf{j} \times \mathbf{B}}{n} + \frac{m}{ne\tau} \mathbf{j}$

$$\mathbf{E} = \frac{1}{ne} \mathbf{j} \times \mathbf{B} + \frac{m}{ne^2\tau} \mathbf{j}$$

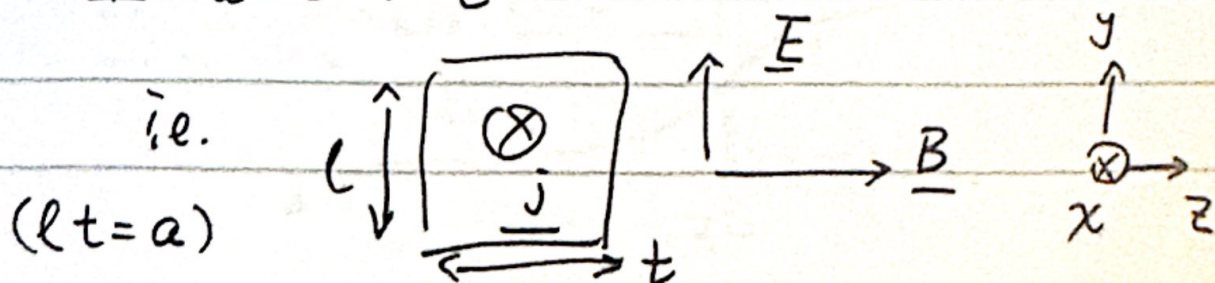
resistivity tensor ρ : $\mathbf{E} = \rho \mathbf{j}$

conductivity tensor σ : $\mathbf{j} = \sigma \mathbf{E}$

$$\sigma_0 = \frac{ne^2\tau}{m} \quad (\text{if } \mathbf{j} \times \mathbf{B} = 0)$$

Hall effect:

- potential difference established across a conductor when a B field is applied $\perp$ to current.



Hall coeff:

$$R_H = \frac{\rho_{yx}}{B_z} = -\frac{1}{ne}$$

Hall voltage:

$$V_H = Ir = I \frac{\rho_{yx} l}{a} = \left[\frac{I B_z R_H}{-t} \right]$$

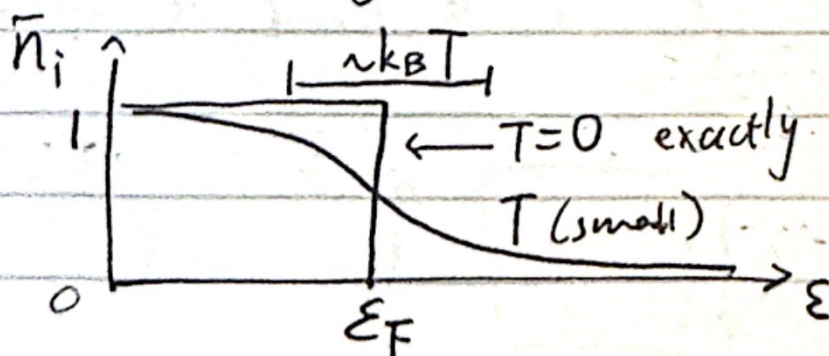
↑
thickness in the
direction of B-field

Lombard theory

$e^- \rightarrow$ fermi gas.

$$\bar{n}_i = \frac{1}{e^{\beta(E-\mu)} + 1} \rightarrow \begin{cases} 1 & E < \mu(T=0) \\ 0 & E > \mu(T=0) \end{cases}$$

Fermi energy $E_F \approx \mu(T=0)$



DOS:

$$g(E)dE = g(k) \frac{dk}{dE} dE$$

$$= \frac{(2s+1)V}{(2\pi)^3} 4\pi k^2 \frac{dk}{dE} dE$$

$$= \frac{(2s+1)V m^{3/2}}{\sqrt{2} \pi^2 \hbar^3} E^{1/2} dE$$

dispersion:

$$E = \frac{\hbar^2 k^2}{2m}$$

$$dE = \frac{\hbar^2}{m} k dk$$

$$k dk = \frac{m}{\hbar^2} dE$$

Determine E_F by $N = \int_0^{E_F} dE g(E)$

$$E_F = \left(\frac{8\pi^2 n}{2s+1} \right)^{2/3} \frac{\hbar^2}{2m}$$

$$T \ll T_F \approx \frac{E_F}{k_B}$$

↑
Fermi temp: at which e^- gas becomes degenerate.

$$k_F = (3\pi^2 n)^{1/3}$$

- Fermi wavevector forms the Fermi surface which divides filled & unfilled states at zero T.

Heat cap of metals

$$\Delta U \propto \Delta N_{\text{excited}} \Delta E \propto (g(E_F) \Delta E) \Delta E$$
$$\propto \frac{N}{E_F} (k_B T) (k_B T)$$

$$\text{So } C_V \propto N k_B \frac{k_B T}{E_F}$$

$$(C_V = \frac{\pi^2}{2} N k_B \frac{k_B T}{E_F} \text{ by Sommerfeld})$$

$$\text{Now } C_V = C_1 T + C_2 T^3 \quad (\text{for metals})$$

↑	↖
electronic	phonon
contribution	contribution
(fermionic, Sommerfeld)	(bosonic, Debye)

12] Crystal & Lattice

lattice: a grid of points defined by a set of 3 primitive lattice vectors $\underline{a}_1, \underline{a}_2, \underline{a}_3$.

Any position $\underline{R} = n_1 \underline{a}_1 + n_2 \underline{a}_2 + n_3 \underline{a}_3$

unit cell: any region of space by which one can repeat by simply translating copies to reconstruct full lattice.

Primitive unit cell: only has 1 lattice point per cell.

Wigner-Seitz cell: the space closer to one lattice point than any other.

Coordination number: # of nearest neighbours of any point in the lattice.

Reciprocal Lattice

$$e^{i \underline{G} \cdot \underline{R}} = 1 \quad (*)$$

formed by points denoted by $\underline{G}$ which

satisfies eqn (*) for all $\underline{R}$ of the 'direct' lattice.

RLV.

$$\underline{G} = h \underline{b}_1 + k \underline{b}_2 + l \underline{b}_3$$

$$\{h, k, l\}$$

↳ Miller indices
(integers)

Thus one can set

$$\underline{a}_i \cdot \underline{b}_j = 2\pi \delta_{ij} \quad (\text{orthogonality relation})$$

$$\text{so } \underline{G} \cdot \underline{R} = 2\pi (hn_1 + kn_2 + ln_3)$$

(no mixed terms)

$$e^{i(\underline{G} \cdot \underline{R})} = 1$$

requires h, k, l all be integers.

$$\text{And } \underline{b}_1 = 2\pi \frac{\underline{a}_2 \times \underline{a}_3}{\underline{a}_1 \cdot (\underline{a}_2 \times \underline{a}_3)} \dots \text{etc.}$$

$\underline{k}$ is conserved (mod $\underline{G}$)
'crystal momentum'

Take F.T. of $\rho(\underline{r}) = \rho(\underline{r} + \underline{R})$

$$\hat{\rho}(\underline{k}) = \sum_{\underline{R}} \int d^3\underline{x} e^{i\underline{k} \cdot (\underline{x} + \underline{R})} \rho(\underline{x} + \underline{R})$$

structure factor

$$= \sum_{\underline{R}} e^{i\underline{k} \cdot \underline{R}} \left[\int d^3\underline{x} e^{i\underline{k} \cdot \underline{x}} \rho(\underline{x}) \right]$$

- Structure factor tells us how unit cell is represented in reciprocal space.

BZ : unit cell in reciprocal space.

(1st BZ $\longleftrightarrow$ Wigner-Seitz cell)

distance between lattice planes:

$$d_{hkl} = \frac{2\pi}{|\underline{G}|} = \frac{2\pi}{\sqrt{h^2|\underline{b}_1|^2 + k^2|\underline{b}_2|^2 + l^2|\underline{b}_3|^2}}$$

Orthorhombic: $|\underline{b}_i| = \frac{2\pi}{|\underline{a}_i|}$

$$\text{so } \frac{1}{d_{hkl}^2} = \frac{h^2}{|\underline{a}_1|^2} + \frac{k^2}{|\underline{a}_2|^2} + \frac{l^2}{|\underline{a}_3|^2}$$

Special case: Cubic $|\underline{a}_i| = a$

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

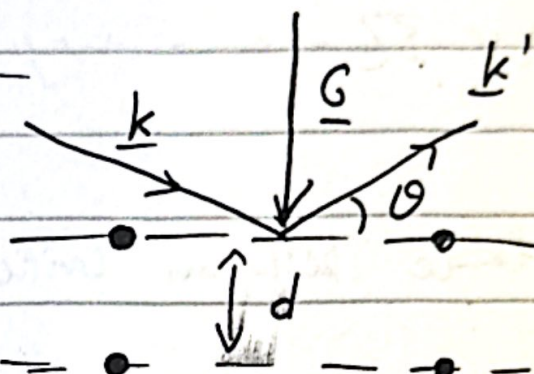
Lane Condition

$$\underline{k} - \underline{k}' = \underline{G} \quad (*)$$

(Conservation of crystal momentum)

Bragg Condition

equiv. to Lane's.



$$\frac{2\pi}{\lambda} (\underline{e}_k - \underline{e}_{k'}) = \underline{G}$$

$$\underline{e}_G \cdot \frac{2\pi}{\lambda} (\underline{e}_k - \underline{e}_{k'}) = \underline{e}_G \cdot \underline{G}$$

$$\frac{2\pi}{\lambda} (2 \sin \theta) = |\underline{G}| \quad \frac{2\pi}{|\underline{G}|} = d$$

$$\underline{2d \sin \theta = \lambda} \quad (**)$$

Constructive interference.

Diffraction intensity

$$\langle \underline{k}' | V | \underline{k} \rangle = \frac{1}{V} \sum_{\underline{R}} e^{i \underline{G} \cdot \underline{R}} \int d^3x e^{i \underline{G} \cdot \underline{x}} V(\underline{x})$$

$$\text{Structure factor} \quad S_{hkl}(\underline{G}) = \int d^3x e^{i \underline{G} \cdot \underline{x}} V(\underline{x})$$

X-Ray Scatt.

roughly $f_j \sim Z_j$

$$S(\underline{G}) \sim \sum_j f_j(\underline{G}) e^{i\underline{G} \cdot \underline{x}_j}$$

$$S_{hkl} = \sum_j f_j e^{2\pi i (hx_j + ky_j + lz_j)}$$

$\uparrow$
form factor (approx.)

Cubic $[0,0,0]$ $S_{hkl} = \sum_j f_j$

BCC $[0,0,0], [\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$

$$S_{hkl} = \sum_j f_j [1 + (-1)^{h+k+l}]$$

($h+k+l=2n, n \in \mathbb{Z}$)

FCC $[0,0,0], [0, \frac{1}{2}, \frac{1}{2}], [\frac{1}{2}, 0, \frac{1}{2}], [\frac{1}{2}, \frac{1}{2}, 0]$

$$S_{hkl} = \sum_j f_j [1 + (-1)^{h+k} + (-1)^{h+l} + (-1)^{k+l}]$$

(allowed only all odd or all even)

Convolutional structure:

$$S_{hkl} = (S_{hkl}^{(\text{lattice})}) (S_{hkl}^{(\text{basis})})$$

X-Ray vs Neutron scatt.

- X-rays $f_j \sim Z_j$, heavy atoms $\checkmark$
Neutrons scatter well with H.
(low Z_j)
- Neutrons short-range
so dependence on G is little.
- Neutrons carry spin (investigate
spin states well)

Debye-Scherrer method (cubic lattice)

- Measure the intensity of concentric
rings on the film (X-ray photographed)
behind the powder sample.

i.e.

(hkl)	$N = h^2 + k^2 + l^2$	Multiplicity
100	1	6 SC
110	2	12 SC, BCC
111	3	8 SC, FCC
200	4	6 SC, BCC, FCC

$$I_{hkl} \propto M_{hkl} / S_{hkl}^2$$

↑

multiplicity factor.

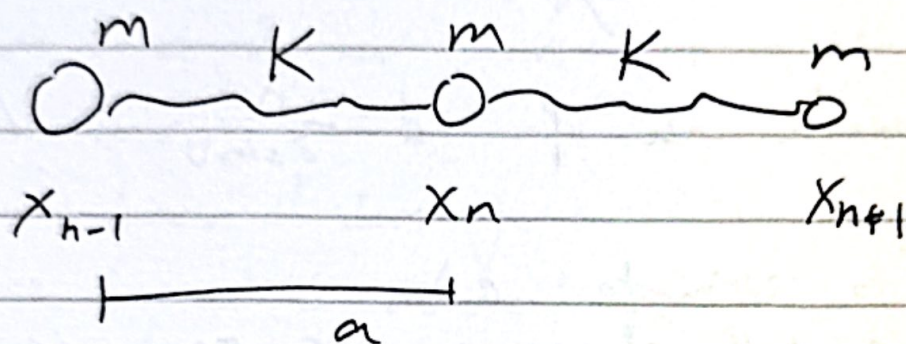
make use of $d = \frac{\lambda}{2 \sin \theta}$ & $d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$

(to estimate a)

& tell if it is SC, FCC, BCC, etc.

[3] Solid Modelling & Band Theory

Monatomic Chain



$$\delta x_n = x_n - x_n^{(eq)}$$

$$\begin{aligned} m\ddot{x}_n &= K(\delta x_{n+1} - \delta x_n) - K(\delta x_n - \delta x_{n-1}) \\ &= K(\delta x_{n+1} + \delta x_{n-1} - 2\delta x_n) \end{aligned}$$

use ansatz $\delta x_n = A e^{i(\omega t - k n a)}$

→ dispersion: $\omega = 2\omega_0 \left| \sin\left(\frac{k a}{2}\right) \right|$

$$\omega_0 = \sqrt{\frac{K}{m}}$$

1st BZ $[-\pi/a, \pi/a]$

since $\delta x_n = \delta x_{n+N}$ (periodic b.c.)

$$e^{i(\omega t - kna)} = e^{i(\omega t - k(N+n)a)} \rightarrow e^{ikNa} = 1$$

$$k = \frac{2\pi p}{Na}$$

$$\# \text{ of modes} = \frac{k_{\text{max}}}{k_{\text{spacing}}} = \frac{\left(\frac{2\pi}{a}\right)}{\left(\frac{2\pi}{Na}\right)} = \underline{\underline{N}}$$

$$V_p = \frac{\omega}{k} = \frac{2\omega_0}{k} \left| \sin \frac{ka}{2} \right|$$

$$V_g = \frac{d\omega}{dk} = a\omega_0 \left| \cos \frac{ka}{2} \right|$$

sound velocity $c_s = \lim_{k \rightarrow 0} V_g \approx \underline{\underline{a\omega_0}}$

DOS: $g(k) dk = 2 \left(\frac{L}{2\pi} \right) dk = \frac{Na}{\pi} dk$

$$g(\omega) d\omega = \frac{Na}{\pi} \left| \frac{dk}{d\omega} \right| d\omega$$

$$= \frac{2N}{\pi \sqrt{(2\omega_0)^2 - \omega^2}}$$

(sub. in
Kos1
wie
 ω & ω_0)

Diatonic chain

$$m_1 \delta \ddot{x}_n = K(\delta y_{n-1} + \delta y_n - 2\delta x_n)$$

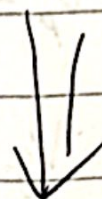
$$m_2 \delta \ddot{y}_n = K(\delta x_{n+1} + \delta x_n - 2\delta y_n)$$

Ansatz: $\delta x_n = A_x e^{i(\omega t - k n a)}$
 $\delta y_n = A_y e^{i(\omega t - k n a)}$

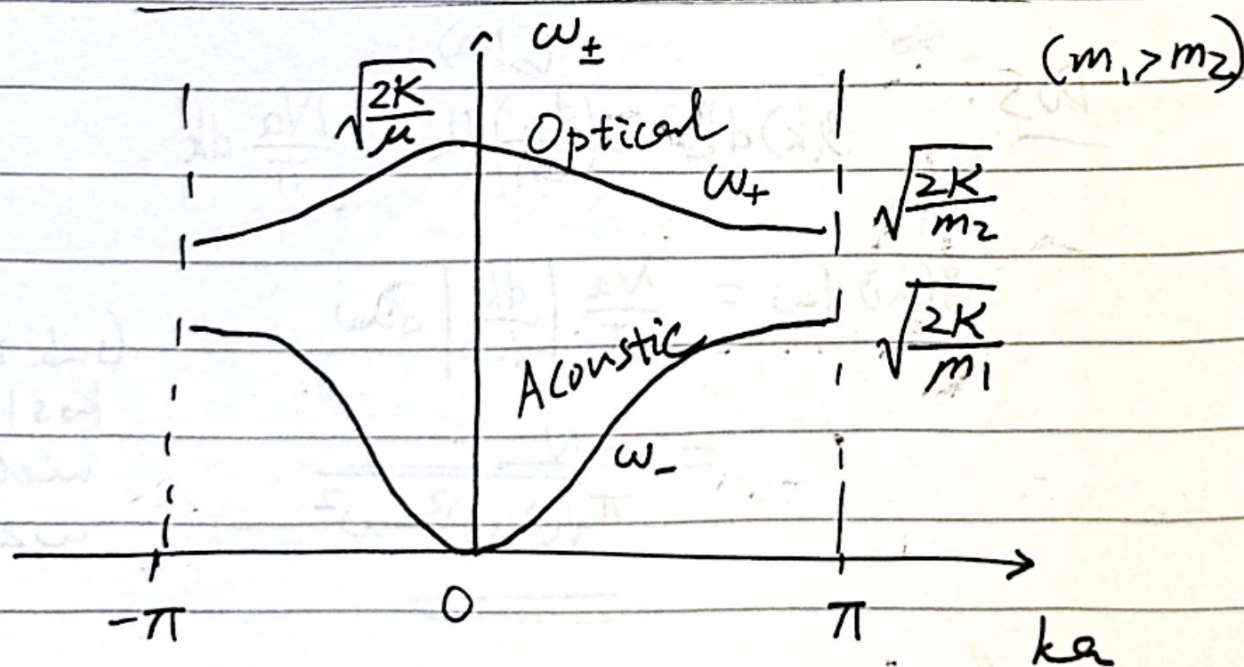
Solve the matrix eqn. $\det = 0$

$$\rightarrow \omega_{\pm}^2 = K \left(\frac{1}{m_1} + \frac{1}{m_2} \right) \pm K \left(\frac{1}{m_1^2} + \frac{1}{m_2^2} + \frac{2}{m_1 m_2} \cos(ka) \right)^{1/2}$$

(2N modes per N unit cells)



$$\omega_{\pm}^2 = \frac{K}{\mu} \left[1 \pm \left(1 - \frac{4\mu}{M} \sin^2\left(\frac{ka}{2}\right) \right)^{1/2} \right]$$



⌞ (R.Z.S)

(E.Z.S)

also possible

Periodic Potentials

Bloch's Theorem

$$\psi_{\underline{k}}^{\alpha}(\underline{r}) = \sum_{\underline{G}} \underbrace{u_{\underline{G}, \underline{k}}^{\alpha}(\underline{r})}_{\substack{\text{periodic} \\ \text{in unit cell}}} e^{i(\underline{G} + \underline{k}) \cdot \underline{r}}$$

- Always look for wave-like solutions for periodic potentials

Tight-binding model

$$\langle n | H | n \rangle = E_0$$

$$\langle n \pm 1 | H | n \rangle = -t$$

$$H = \sum_n (E_0 |n\rangle\langle n| - t(|n\rangle\langle n-1| + |n\rangle\langle n+1|))$$

$$\text{Guided } |\psi\rangle = \sum_n \phi_n |n\rangle$$

$$H|\psi\rangle = E|\psi\rangle$$

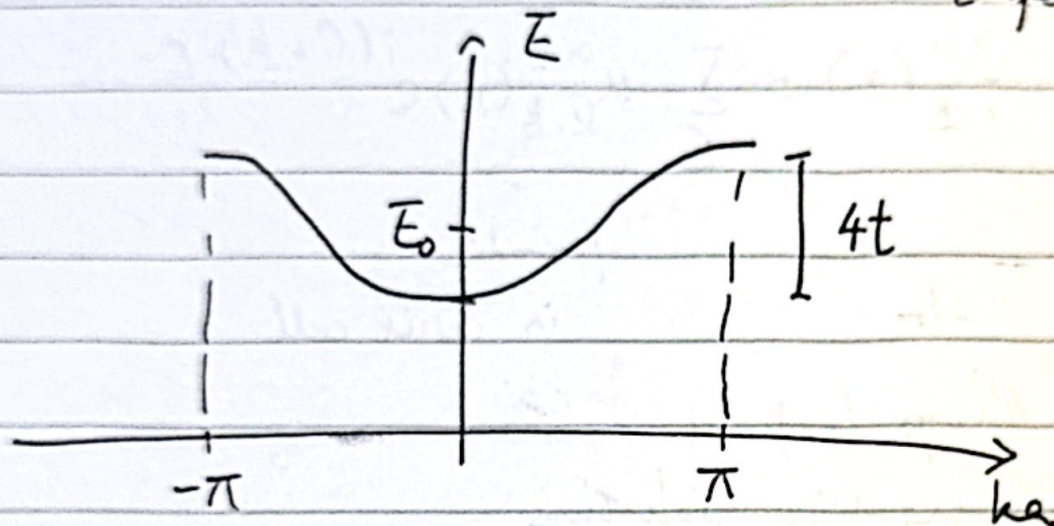
↓

$$E_0 \phi_n - t \phi_{n-1} - t \phi_{n+1} = E \phi_n$$

Use ansatz $\psi_n = \frac{e^{i(ut - kra)}}{\sqrt{N}}$

Set $\bar{E}(k) = \bar{E}_0 - 2t \cos(ka)$

dispersion



DOS: $g(k) dk = 2(2s+1) \frac{L}{2\pi} dk$

$g(\bar{E}) d\bar{E} = 2(2s+1) \frac{L}{2\pi} \left| \frac{dk}{d\bar{E}} \right| d\bar{E}$

$(s = \frac{1}{2}) \quad = \frac{2N}{\pi \sqrt{(2t)^2 - (E - E_0)^2}} dE$

Newly free e^- model

$\langle \underline{k} | H | \underline{k} \rangle = E_0(\underline{k}) + V_0$

$\langle \underline{k}' | H | \underline{k}' \rangle = E_0(\underline{k}') + V_0$

$\langle \underline{k} | H | \underline{k}' \rangle = V_0$

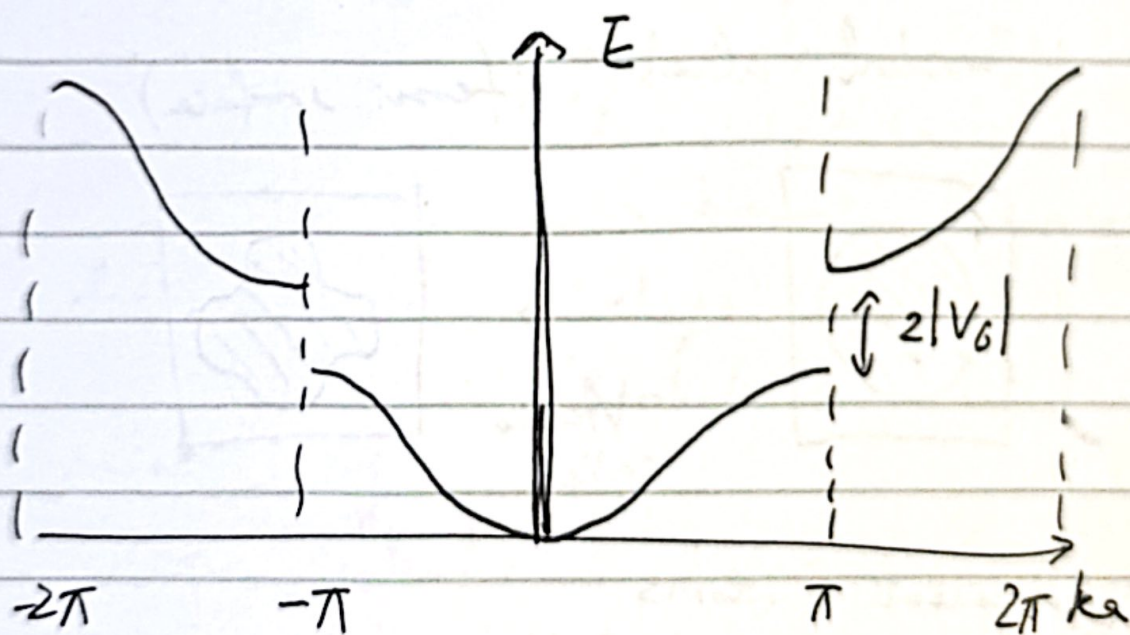
$(= \langle \underline{k}' | H | \underline{k} \rangle)$

Schrödinger's eqn: basis: $\{|k\rangle, |k'\rangle\}$

$$\begin{bmatrix} E_0(k) + V_0 - E & V_G \\ V_G & E_0(k') + V_0 - E \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

At BZ boundary $k = k'$

$$E_{\pm}(k) = \frac{\hbar^2 k^2}{2m} + V_0 \pm |V_G|$$



Periodic potential $\rightarrow$ discrete energy bands
& band gaps.
(in dispersion)

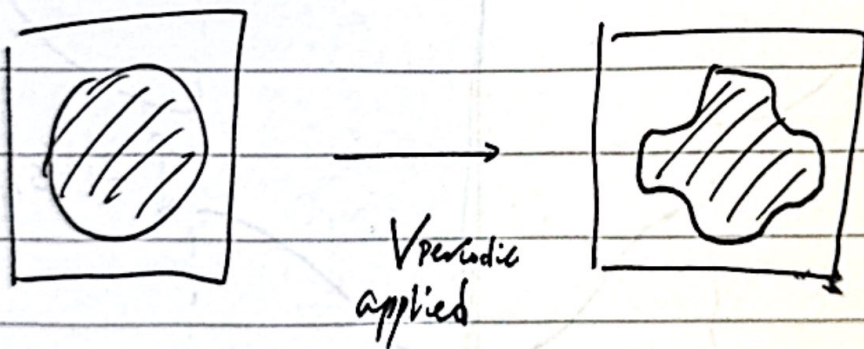
Band Theory & Semiconductors

Conductors: partially filled band(s)

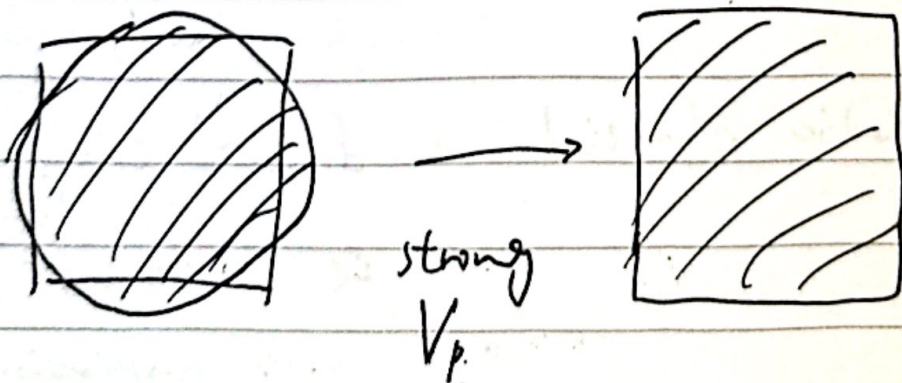
Insulators: fully filled bands

Semiconductors: fully filled bands
with small band gap
($\leq 4\text{eV}$)

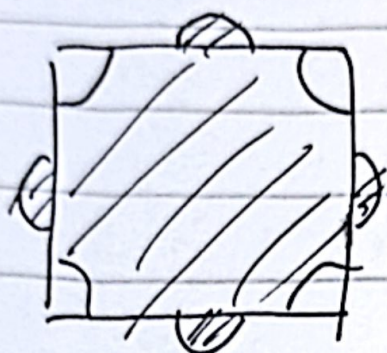
Monovalent atoms (fermi surface)



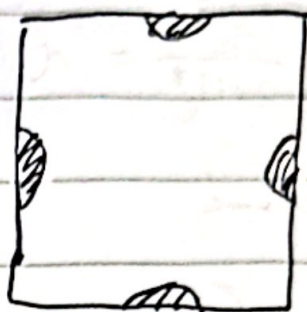
Divalent atoms



Moderate V_p



Extended



Reduced

Direct transitions: do not change k .
(photons alone cannot have indirect transitions, as momentum is not conserved)

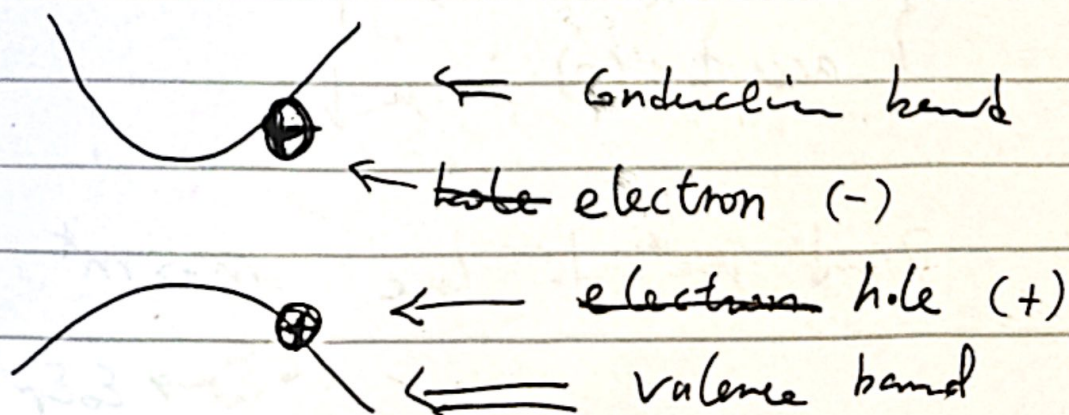
Indirect transitions: change value of k .
(involve lattice phonons)

Band theory

ignores Coulomb int., can fail

in describing Mott insulators

(where e^-e^- repulsion dominates)



cond. band

$$E = E_{\min} + \alpha |\underline{k} - \underline{k}_{\min}|^2$$

$$\frac{\hbar^2}{2m_e^*} = \alpha$$

val. band

$$E = E_{\max} - \alpha |\underline{k} - \underline{k}_{\min}|^2$$

$$\frac{\hbar^2}{2m_h^*} = \alpha$$

$$\underline{k}_e = -\underline{k}_h$$

Carrier mobility $\mu = \frac{|\underline{v}|}{|\underline{E}|} = \left| \frac{e\tau}{m^*} \right|$

• how easily carriers move around.

Intrinsic: $n = p$

Impurities/Doping $\rightarrow n \neq p$

$$\begin{cases} \text{donors (n)} \Rightarrow n \uparrow \\ \text{acceptors (p)} \Rightarrow n \downarrow \end{cases}$$

Doped materials have $m \rightarrow m^*$
 $\epsilon \rightarrow \epsilon_0 \epsilon_r$

$$R^{\text{eff}} = R \left(\frac{m^*}{m} \frac{1}{\epsilon_r^2} \right)$$

$$a_0^{\text{eff}} = a_0 \left(\frac{m_e}{m_e^*} \epsilon_r \right)$$

Law of Mass Action

starting with

$$g(E) dE = (2s+1) \frac{V}{(2\pi)^3} 4\pi k^2 \left| \frac{dk}{dE} \right| dE \propto E^{1/2} dE$$

$$\Rightarrow \begin{cases} g_c(E \geq E_c) \propto (E - E_c)^{1/2} \\ g_v(E \leq E_v) \propto (E_v - E)^{1/2} \end{cases}$$

$$\bar{n}_c = \frac{1}{e^{\beta(E-\mu)} + 1} \approx \frac{e^{-\beta(E-\mu)}}{1} \quad \begin{matrix} |\beta(E-\mu)| \gg 1 \\ \beta(E-\mu) > 0 \end{matrix}$$

$$\bar{n}_v = 1 - \frac{1}{e^{\beta(E-\mu)} + 1} \approx \frac{e^{\beta(E-\mu)}}{1} \quad \begin{matrix} |\beta(E-\mu)| \gg 1 \\ \beta(E-\mu) < 0 \end{matrix}$$

$$\text{set } n \propto (m_e^* k_B T)^{3/2} e^{-\beta(E_c - \mu)}$$

$$p \propto (m_h^* k_B T)^{3/2} e^{-\beta(\mu - E_v)}$$

$$\text{so } np = \frac{1}{2} \left(\frac{k_B T}{\pi \hbar^2} \right)^3 (m_e^* m_h^*)^{3/2} e^{-\beta E_g}$$

band gap energy: ($E_g = E_c - E_v$)

Independent of μ .

Taking ratios of n & p ,

(assuming intrinsic $n=p$)

$$\rightarrow \mu = \frac{1}{2} (E_c + E_v) + \frac{3}{4} k_B T \ln \left(\frac{m_h^*}{m_e^*} \right)$$

$\rightarrow$ At $T=0$, μ is halfway between.

Poping $D=n-p$, law of mass action still holds.

3 Regimes of T (doped materials)

low T

high T

freeze-out

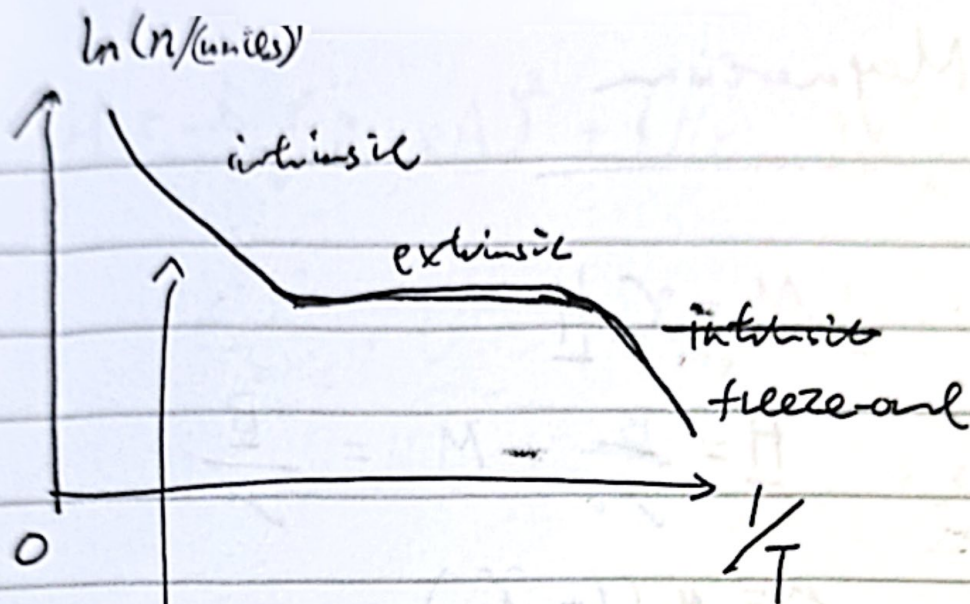
extrinsic

intrinsic

carriers
atom-bound.

donors
ionised
(plateau)

excite intrinsic
electron-hole
pairs.



gradient = $-\frac{E_g}{2k_B}$

(4) Magnetism

Useful: $\underline{M} = \chi_m \underline{H}$

$$\underline{H} = \frac{\underline{B}}{\mu_0} - \underline{M} = \frac{\underline{B}}{\mu}$$

$$\mu = \mu_0 (1 + \chi_m)$$

Hund's Rules (for ground state)

- ① e^- tries to align spin with one another.

(antisymmetric spatial wavefunction has lower energies)

- ② maximise total L .

- ③ Total $J = |L \pm S|$

more than half filled (+)

less than half filled (-)

Under external B-field:

Paramagnetism $J \neq 0$

Diamagnetism $J = 0$ (otherwise too small)

$$H = \frac{1}{2m_e} (\underline{p} + e\underline{A})^2 + U(r) + g_s \frac{\mu_B}{\hbar} \underline{B} \cdot \underline{S}$$

$$= \underbrace{\frac{\underline{p}^2}{2m_e} + U(r)}_{H_{CF}} + \frac{e}{2m_e} \underline{B} \cdot \underline{L} + g_s \frac{\mu_B}{\hbar} \underline{B} \cdot \underline{S} + \frac{e^2}{2m_e} \frac{1}{4} |\underline{B} \times \underline{r}|^2$$

Using $\underline{A} = \frac{1}{2} \underline{B} \times \underline{r}$ & $\underline{r} \times \underline{p} = \underline{L}$

$$\mu_B = \frac{e\hbar}{2m_e}$$

$$H = H_{CF} + \underbrace{\frac{\mu_B}{\hbar} (\underline{L} + g_s \underline{S}) \cdot \underline{B}}_{\text{Paramagnetism}} + \underbrace{\frac{e^2}{2m_e} \frac{1}{4} |\underline{B} \times \underline{r}|^2}_{\text{Diamagnetism}}$$

$$(\underline{\mu} = - \frac{\mu_B}{\hbar} (\underline{L} + g_s \underline{S})) \downarrow$$

tends to
decrease
E of
system

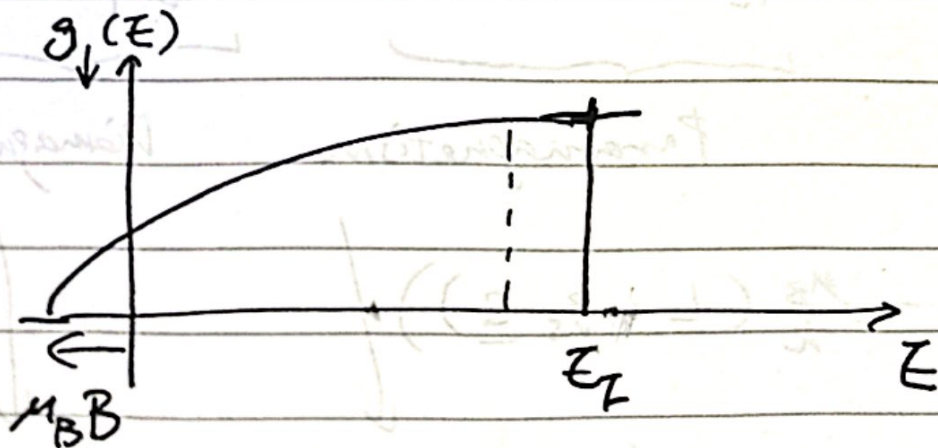
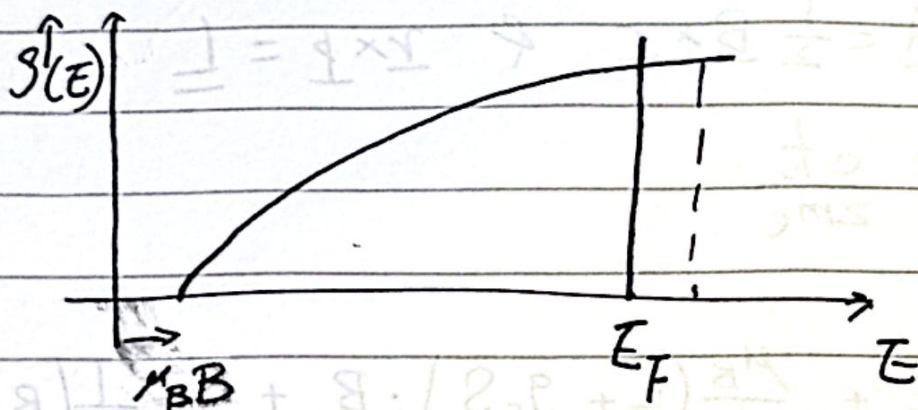
$\downarrow$
tends to
increase
E of
system.

Pauli Paramagnetism

(Fermi gas
response
to external B)

$$\Delta E \sim \mu_B B$$

$$M = -\frac{1}{V} \frac{dE}{dB} \sim -(\Delta \#_{\text{up}} - \Delta \#_{\text{down}}) \frac{\mu_B}{V}$$



$$M = -\left[\left(-\frac{g(E_F)}{2} \mu_B B \right) - \left(+\frac{g(E_F)}{2} \mu_B B \right) \right] \mu_B$$
$$= g(E_F) \mu_B^2 B$$

$$\chi_{\text{Pauli}} = \frac{dM}{dH} = \mu_0 \frac{dM}{dB} = \boxed{\mu_0 \mu_B^2 g(E_F)}$$

Curie Paramagnetism (free-spin in atom)

$$\Delta E = g_J \mu_B B M_J$$

$$\text{so } Z_1 = \sum_{M_J=-J}^J e^{\underbrace{-(\beta \mu_B B g_J) M_J}_x}$$

$$= e^{-xJ} + e^{-x(J-1)} + \dots + e^{x(J-1)} + e^{xJ}$$

$$= e^{-xJ} (1 + e^x + \dots + e^{2xJ})$$

$$= e^{-xJ} \frac{1 - e^{(2J+1)x}}{1 - e^x}$$

$$= \frac{e^{-xJ} - e^{x(J+1)}}{1 - e^x} = \frac{\sinh\left[(J+\frac{1}{2})x\right]}{\sinh(\frac{1}{2}x)}$$

$$F = -k_B T \ln(Z_1^N)$$

$$= -N k_B T \ln \left(\frac{\sinh\left[(J+\frac{1}{2})x\right]}{\sinh(\frac{1}{2}x)} \right)$$

$$M = -\frac{1}{V} \left(\frac{\partial F}{\partial B} \right)_T = n \mu_B g_J J \underbrace{B_J(x)}$$

Brillouin function.

$B_J \rightarrow 1$ if $x \gg 1$ strong alignment

$B_J \rightarrow 0$ if $x \ll 1$ weak alignment

Curie's law $\rightarrow \chi_{\text{Curie}} \propto n \mu_B g_J J \frac{\partial B_J}{\partial H}$

$$\chi_{\text{Curie}} \propto n \mu_B g_J J \left(\mu_0 \frac{\partial \chi}{\partial B} \right) \quad (x = g_J \mu_B B \beta)$$

$$\propto n \mu_0 (g_J \mu_B)^2 J \beta$$

$$\propto n \mu_0 (g_J \mu_B)^2$$

$$k_B T$$

actually

$$\chi_{\text{Curie}} \propto \frac{1}{T}$$

$$\chi_{\text{Curie}} = \frac{n \mu_0 (g_J \mu_B)^2 J(J+1)}{3 k_B T}$$

• if J non-zero, Curie param. dominates at very low temperatures.

Larmor Diamagnetism

$$\delta H_d = \frac{e^2}{8 m_e} |\underline{B} \times \underline{r}|^2 = \frac{e^2 B^2}{8 m_e} \langle x^2 + y^2 \rangle$$

$$= \frac{e^2 B^2}{8 m_e} \left(\frac{2}{3} \langle r^2 \rangle \right)$$

$$\delta E_J = \frac{e^2 \hbar^2}{12 m_e} \langle r^2 \rangle$$

$$m = - \frac{\partial \delta E_J}{\partial B} = - \frac{e^2}{6 m_e} \langle r^2 \rangle B \quad (\text{per } e^-)$$

$$\chi_{\text{Larmor}} = + \mu_0 \beta \frac{\partial m}{\partial B} = \underline{\underline{- \frac{\mu_0 \beta e^2 \langle r^2 \rangle}{6 m_e}}}$$

Spontaneous Magnetism

Domains → responsible.

Hysteresis loop (history dependent)

Weiss Mean Field Theory

Heisenberg model:

$$H = - \frac{1}{2} \sum_{\langle i, j \rangle} J(\underline{S}_i \cdot \underline{S}_j) + g_s \mu_B \underline{B} \cdot \sum_i \underline{S}_i$$

For a single site i : $H_i = [g_s \mu_B \underline{B} - J \sum_j \underline{S}_j] \cdot \underline{S}_i$

$$H_i = g_s \mu_B \left(B - \underbrace{\frac{J}{g_s \mu_B} \sum_j S_j}_{\langle B_{\text{eff}} \rangle} \right) S_i$$

Eigenvals: ~~$H_i = g$~~ $E_i = \pm \frac{g_s \mu_B \langle B_{\text{eff}} \rangle}{2}$

S.M.

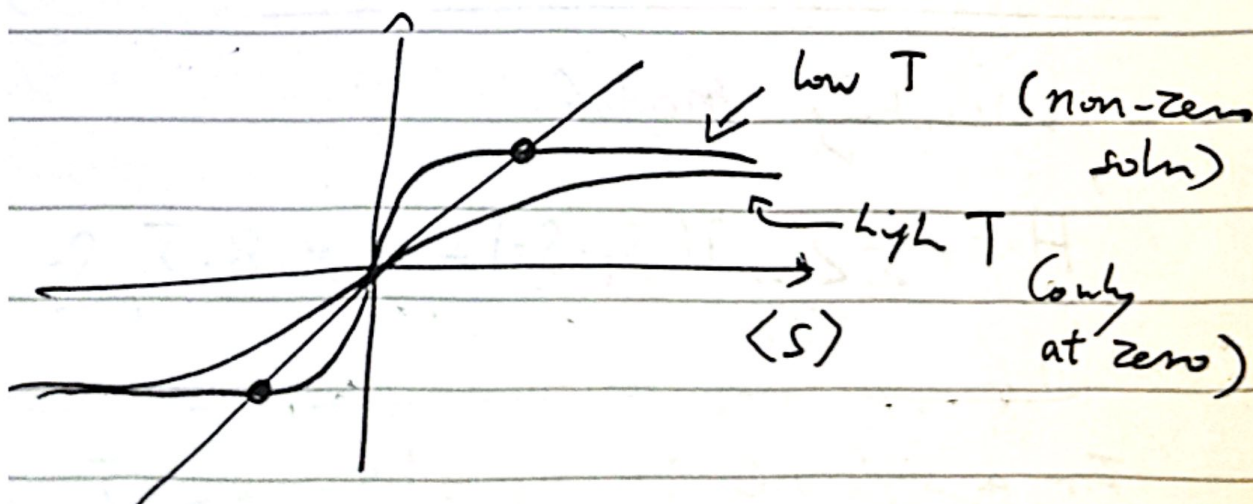
$$\rightarrow F = -k_B T \ln \left[2 \cosh \left(\frac{1}{2} g_s \mu_B \beta \langle B_{\text{eff}} \rangle \right) \right]$$

$$\langle S \rangle = \frac{\partial F}{\partial \langle g_s \mu_B B_{\text{eff}} \rangle}$$

$$= -\frac{1}{2} \tanh \left(\frac{1}{2} g_s \mu_B \beta \langle B_{\text{eff}} \rangle \right)$$

$$= -\frac{1}{2} \tanh \left(\frac{1}{2} \beta (g_s \mu_B B - J_z \langle S \rangle) \right)$$

$B=0$ case $\langle S \rangle = \frac{1}{2} \tanh \left(\frac{1}{2} \beta J_z \langle S \rangle \right)$



transition at $\tanh x \approx x$

$$k_B T_c = \frac{Jz}{4}$$

Curie temperature.

Expand $\langle S \rangle$ by $\tanh x \approx x$ (high T)

$$\rho \langle S \rangle \ll 1$$

$$\langle S \rangle \approx -\frac{1}{4} \frac{g_s \mu_B B}{k_B T (1 - T_c/T)}$$

$$M = -\frac{g_s \mu_B}{V_{uc}} \langle S \rangle$$

$$\chi_M = \mu_0 \frac{dM}{dB} = \left[\frac{\mu_0 (g_s \mu_B)^2}{4k_B T (1 - T_c/T)} \right]$$

Curie-Weiss susceptibility

$$\chi_{\text{weiss}} = \frac{\chi_{\text{curie}}}{(1 - T_c/T)} \quad (\text{Curie-Weiss Law})$$

low T $\longleftrightarrow$ high T
ferromag. $\sim T_c$ paramag.

Dynamics of holes vs electrons

$$\left\{ \begin{array}{l} \underline{k}_h = -\underline{k}_e \\ E_h = -E_e \\ \underline{v}_h = \underline{v}_e \\ m_h^* = -m_e^* \end{array} \right.$$

$$\left\{ \begin{array}{l} \hbar \frac{d\underline{k}_h}{dt} = e(\underline{E} + \frac{1}{c} \underline{v}_h \times \underline{B}) \\ \hbar \frac{d\underline{k}_e}{dt} = -e(\underline{E} + \frac{1}{c} \underline{v}_e \times \underline{B}) \end{array} \right.$$

	# atoms	# nearest neighbors (Z)
SC	1	6
BCC	2	8
FCC	4	12